Enhancing molecular property prediction via transformer with dual graph representation

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Enhancing molecular property prediction via transformer with dual graph representation

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Abstract

Accurate prediction of molecular properties is central to advancing chemistry, materials science, and drug discovery. Attention-based transformers have recently emerged as an effective infrastructure for learning on molecular graphs, yet their full potential depends on representations that capture the rich topology and structure of molecules. To this end, we propose the dual graph transformer (DGT), a self-attention architecture that jointly models atom and bond graphs to achieve comprehensive molecular representation. DGT facilitates the fusion of atom and bond features, graph topology and structure, and three-dimensional information within the self-attention module to learn an effective molecular representation. We benchmark DGT across a range of datasets for molecular property prediction, showing that it considerably outperforms the current state of the art. DGT demonstrates performance contributions from its dual graph representation, relative positional and structural encodings, and spatial information incorporation, while also offering interpretability at the molecular structural level. We envision DGT advancing molecular machine learning by improving both the prediction accuracy and interpretability of molecular properties.

Main

Molecular science sits as a cornerstone of many application domains, as it underpins our understanding and manipulation of matter at the fundamental level[1,2]. Within molecular sciences, accurate prediction of molecular properties plays a critical role in exploring chemical spaces across versatile domains where wet-lab experiments and first-principles calculations are often costly and time-consuming[3,4]. Deep learning methods, which bypass the need for hand-crafted feature design through automatic feature extraction, offer a promising avenue toward high-throughput screening of molecules for desired properties, such as chemical reactivity[5], solubility parameters[6], HOMO–LUMO gaps[7], and binding affinities to proteins[8]. Beyond accelerating molecular screening, these methods also provide new insights into the underlying factors that drive molecular behaviours of great interest.

At the core of molecular property prediction lies the approach for representing molecules, which is closely connected to model architectures and critically influences learning performance[9]. A wide spectrum of molecular representations has been explored: Fingerprints and descriptors encode molecules as fixed-length vectors by summarising substructures or cheminformatics-derived features[10]; string-based representations, such as the simplified molecular-input line-entry system, tokenise molecules as sequence data which are readily handled by recurrent neural networks and large language models[11]; and graph-based representations capture atom and bond features alongside connectivity patterns, providing a more faithful depiction of molecular structures[12]. Such molecular graphs support graph neural networks (GNNs) to directly learn representations from molecular structures instead of pre-computed features, attracting growing interest. However, the message-passing mechanism underlying GNNs comes with challenges of over-smoothing and limited k-hop neighbourhoods, constraining the expressivity and receptive field of node features[13,14]. Meanwhile, the attention mechanism achieves substantial success in modelling sequence data, motivating its application to molecular representation learning[15]. The graph transformer, which harnesses the self-attention mechanism to enable global dynamic node–node interactions regardless of their graph distances, is surging recently[16].

Employing the self-attention mechanism on molecules requires the comprehensive encoding of molecular information, including atom and bond features, graph topology and structure, and three-dimensional (3D) information if available, in the query-key-value operation[17,18]. Graph-specific encodings—including relative positional encoding (RPE)[19] and ring structural encoding (RSE)[20]—can be incorporated into the graph transformer layer, along with edge features, to capture complex dependencies within molecules[21]. In some graph transformer implementations, edge features are also aggregated as characterisation of local neighbourhoods for updating node features before the attention calculation[22]. Despite that molecules are often intuitively depicted as topological graphs with atoms as nodes, reinterpreting bonds as nodes in molecular graphs can provide an alternative perspective to enrich molecular representation in the attention mechanism and thereby promote property prediction[23].

To this end, we propose the dual graph transformer (DGT), a novel deep learning architecture that leverages the self-attention mechanism for enhancing molecular property prediction by integrating the atom graph with its bond counterpart. Atom and bond features are mutually fused into the attention score matrices of their counterparts. RPE and RSE are incorporated into the attention matrices of both atom and bond graphs. DGT also enables encoding bond lengths, atom–atom distances, and bond–bond angles into the self-attention mechanism for the optional adoption of 3D information of different fidelities. DGT is benchmarked across 10 datasets and 58 tasks for molecular property prediction and considerably outperforms the current state of the art. Further performance investigation validates contributions of dual graph representation, graph-specific encodings, spatial structure incorporation, and model pretraining to molecular property prediction. We also demonstrate that the attention-based interpretation can reveal crucial substructures for molecular properties on the dual graph representation. DGT presents as a powerful deep learning architecture for effective molecular representation learning to advance prediction and understanding of molecular properties.

Results

Overview of DGT

The proposed DGT is illustrated in Fig. 1, encompassing stages of graph representation, molecule encoding, stacked graph transformer layers, and a final readout module for molecular property prediction. The bond graph representation is derived by reinterpreting the intuitive atom graph with the line graph transform, which exchanges roles of atoms and bonds in the topological structure. This provides an alternative perspective to enrich molecular features, thereby enhancing the property prediction performance of deep learning approaches. As the exemplified phenyl formate in Fig. 1a, carbon and oxygen atoms are regarded as edges in the bond graph representation, depicting connections of single, double, and aromatic bonds. The highlighted *ipso* carbon in the atom graph corresponds to the three highlighted edges, sharing the same feature vector, in the bond graph.

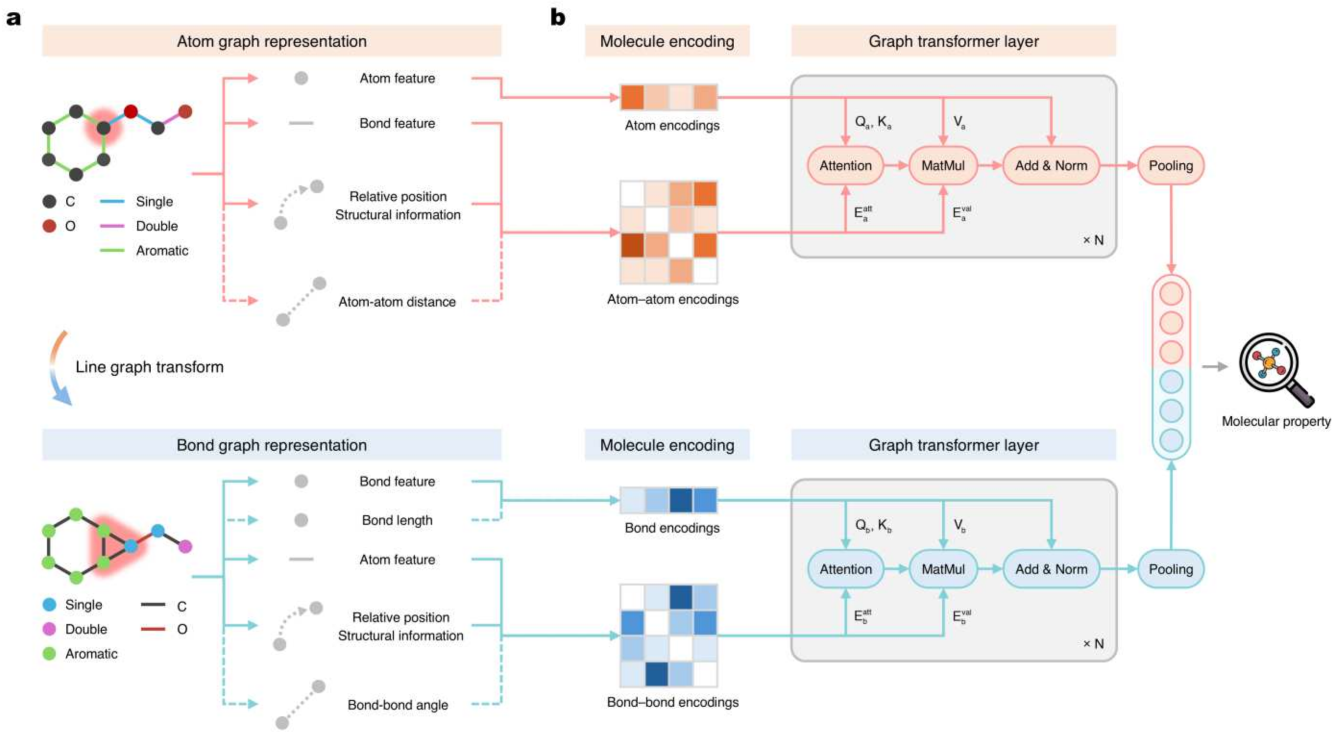

Fig. 1: An illustration of DGT. a, Atom and bond graph representations of molecules. Positions of atoms and bonds are exchanged in the dual graph representation, accommodating atom and bond features, relative position, and structural information. Bond length, atom–atom distance, and bond–bond angle can also be integrated into DGT when available. **b**, Molecule encodings and graph transformer structure. DGT encodes molecules into atom and bond vectors, atom–atom matrices, and bond–bond matrices. Graph transformer layers are stacked to conduct the query-key-value operation, followed by the pooling layer for predicting molecular properties.

Such a dual graph representation realises the encoding of comprehensive molecular information—including atom and bond features, graph topology, specific structures, and 3D information if available—into deep learning architectures for molecular learning tasks. In detail, atom and bond features are represented as node vectors in their respective graphs and as edge vectors in their mutual graphs. Shortest path distance encoding (SPDE) and random walk positional encoding (RWPE) are employed as the RPE of nodes; RSE is added to deliver ring information, given its importance for many molecular properties. Together, these graph-specific encodings are combined with edge features to form node–node matrices as pairwise features for both atom and bond graphs. Additionally, this dual graph representation accommodates diverse 3D information: Bond lengths can be embedded within bond features; atom–atom distances and bond–bond angles can be fit into the pairwise feature matrices of atom and bond graphs, respectively.

The graph transformer layer extends the conventional query-key-value operation by merging the pairwise feature matrices from atom and bond graphs into their respective attention score matrices and passing messages, allowing DGT to learn both local and global dependencies for atoms and bonds (Fig. 1b). In addition, the pairwise feature matrices are also utilised to update node vectors within the self-attention calculation, providing an effective means to enhance node messages, as demonstrated in recent studies[24,25]. Batch normalisation is applied in each graph transformer layer to stabilise training. Details of the graph transformer layer are provided in the Methods section. After stacked graph transformer layers, the global average pooling is employed as a readout function, followed by the concatenation operation of outputs of both graph representations and fully connected layers for molecular property prediction.

Molecular property prediction benchmark

We conducted a comprehensive benchmark study of DGT on molecular property prediction across 10 datasets and 58 tasks, spanning four domains: physiology, biophysics, physical chemistry, and quantum mechanics. DGT was evaluated against nine state-of-the-art machine learning models to demonstrate its superior performance in learning on molecules. As a classical baseline, the random forest (RF) model was built using ECFP4 fingerprints as input features[26]. Attentive FP[27] and D-MPNN[28] served as representative GNN baselines. In addition, pretrained GNN[29], GraphMVP[30], MolCLR[31], and MoleBERT[32] were included as GNN models leveraging large-scale molecular corpora for pretraining followed by task-specific fine-tuning. GROVER was implemented as a graph transformer framework capturing rich structural and semantic information within molecular graphs[33]. UniMol represented as a transformer-based 3D geometric deep learning model for molecular property prediction[34].

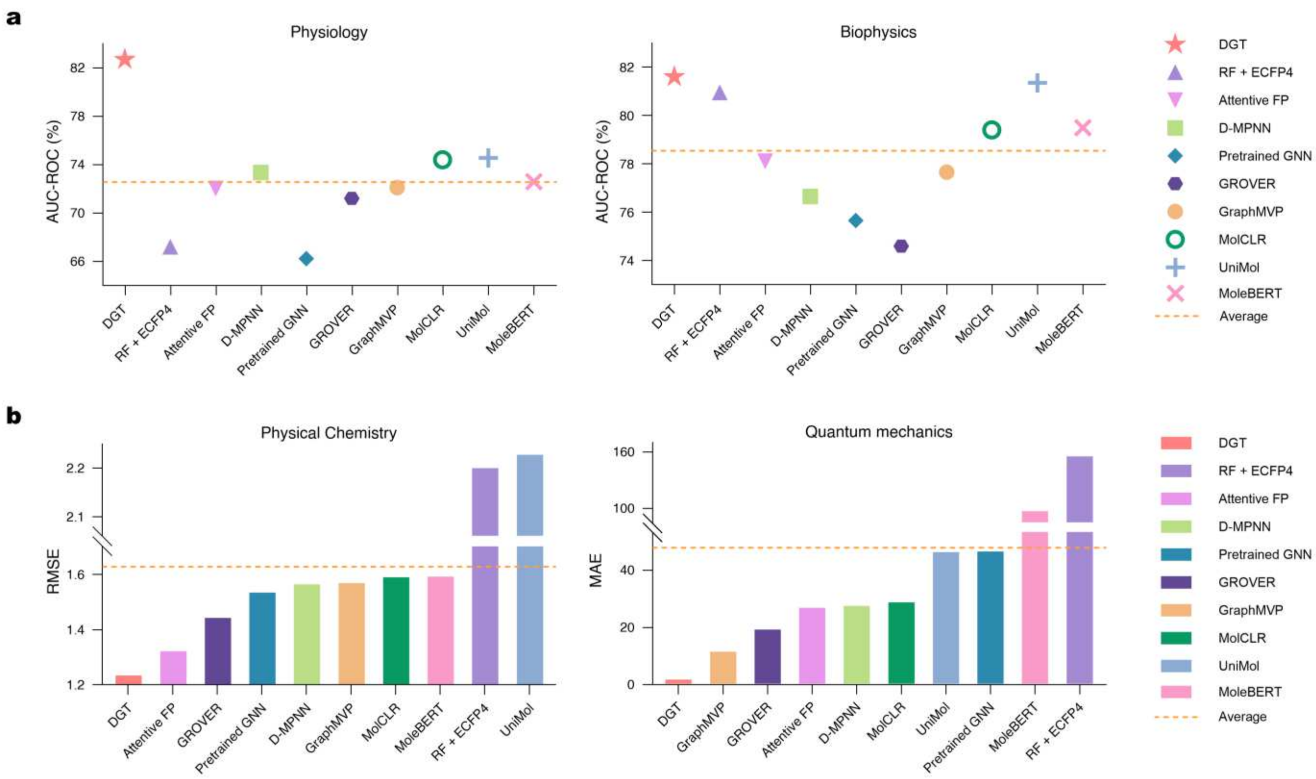

Fig. 2: Benchmarking DGT against baselines across four domains. a, Performance comparison of DGT against baselines for predicting categorical molecular properties in physiology and biophysics domains. **b**, Performance comparison of DGT against baselines for predicting numerical

molecular properties in physical chemistry and quantum mechanics domains.

DGT outperformed all baseline models on classification tasks in the physiology and biophysics domains, as illustrated in Fig. 2a. Within the physiology domain, DGT displayed a substantial uplift in the area under the receiver operating characteristic curve (AUC-ROC), delivering a profound 8.08% improvement over the second-best model's 74.55% and vastly exceeding the average AUC-ROC of 72.59%. Concurrently, in the biophysics domain, DGT likewise attained the top performance, registering an AUC-ROC of 81.60% and performing competitively with the pretrained transformer-based UniMol model. These collective results affirm DGT's broad applicability and high accuracy in predicting molecular properties relevant to drug discovery, including pharmacokinetics, toxicity, bioactivity, and potential adverse effects.

In the physical chemistry and quantum mechanics domains, DGT also observed significant advantages over all baselines (Fig. 2b). For physical chemistry tasks, DGT suggested a low root mean square error (RMSE) of 1.234, representing a notable 6.66% performance advancement over the next best model. In comparison, GNN baselines with pretraining reached a minimum RMSE of 1.443, and the RMSE values of RF and UniMol were even larger than 2.2. In the quantum mechanics domain, DGT outstood all other models, achieving a markedly low mean absolute error (MAE) of 1.545, which is an order of magnitude lower than that of the second-best model, GraphMVP. These results unveil the exceptional ability of DGT to precisely predict numerical physicochemical and quantum mechanical properties for molecules.

DGT surpasses other models by providing a rational approach to molecular representation and intimately coupling molecule encodings with the self-attention mechanism in the context of graph-structured molecular data. Unlike natural languages, where attention was initially devised to model sequential word dependencies, molecular graphs pose unique challenges arising from their complex topological structures and the absence of an inherent sequential order. Consequently, executing the attention mechanism in molecular machine learning necessitates specialised adaptations to capture both local and long-range interactions among atoms and bonds. The graph transformer layer addresses this by enabling nodes to aggregate information directly from the entire graph, rather than being restricted to their *k*-hop neighbourhoods, while embedding graph-specific encodings into the attention score matrix, including topological and structural information.

Performance investigation of DGT

We investigated the performance of DGT through comparative analyses focused on the prediction of highest ($\varepsilon_{\mathrm{HOMO}}$) and lowest ($\varepsilon_{\mathrm{LUMO}}$) unoccupied molecular orbital energies, two pivotal electronic properties governing reactivity and stability, using the QM9 dataset[26]. As shown in Fig. 3a, both atom and bond graph representations demonstrably contributed to predicting $\varepsilon_{\mathrm{HOMO}}$ and $\varepsilon_{\mathrm{LUMO}}$. Removing the bond graph representation from DGT led to an average MAE increment of 2.7%, while omitting the atom graph representation resulted in a more substantial 5.3% increment. Similarly, RPE and RSE, which were included in the self-attention calculation as structural and topological information, proved critical (Fig. 3b). Their exclusion produced average MAE increases of 2.8% and 6.8%, respectively.

Collectively, DGT revealed an average performance degradation of 10.2% when both RPE and RSE were omitted. These findings substantiate the rationale of the dual graph representation to integrate comprehensive molecular information, underscoring the synergistic effects of DGT components in enhancing graph transformer's predictive power for molecular properties.

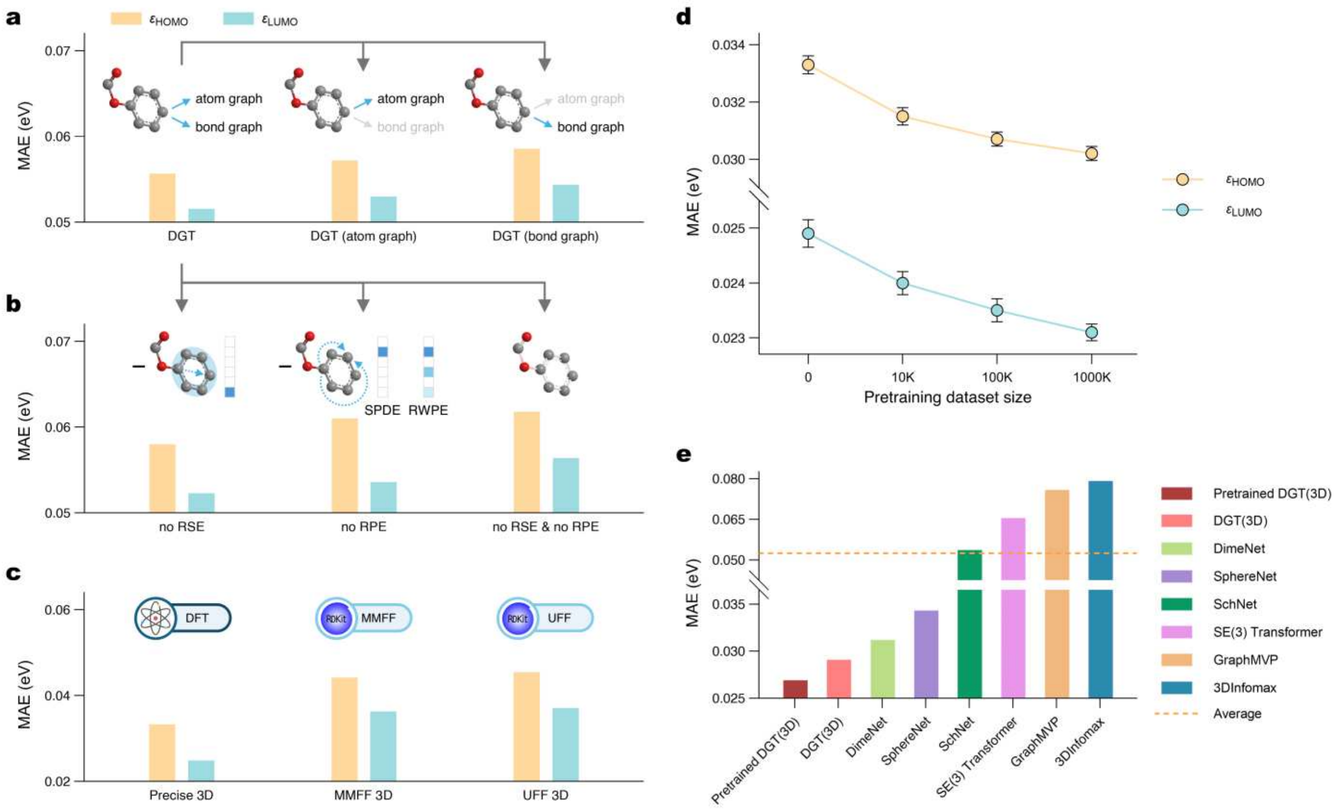

Fig. 3: Performance investigation of DGT and 3D information incorporation. HOMO and LUMO energy prediction tasks from the QM9 dataset are used for demonstration. The 3D information-augmented DGT, denoted as DGT(3D), undergoes pretraining on molecules from the PCQM4Mv2 dataset targeting the HOMO–LUMO energy gap. **a**, Ablation study evaluating the dual graph representation. **b**, Ablation study of graph-specific encodings. **c**, Effect of incorporating 3D information at different fidelity levels on DGT's predictive performance. **d**, Performance of DGT(3D) as a function of pretraining dataset scale. **e**, Comparative assessment of DGT(3D) against state-of-the-art models for 3D molecular learning using the average prediction MAE of HOMO and LUMO energies.

Fig. 3c elucidates the predictive capabilities of DGT when augmented with 3D structural information of varying fidelities. We investigated DGT's performance using precise spatial coordinates derived from Density Functional Theory (DFT) calculations for constructing the QM9 dataset, alongside approximate geometries generated using the Merck Molecular Force Field (MMFF) and the Universal Force Field (UFF) implemented in RDKit[35]. The results demonstrated DGT's ability to utilise 3D information across different fidelities for improving ε_{HOMO} and ε_{LUMO} predictions. Specifically, the precise DFT-derived 3D information yielded a significant average MAE reduction of 48.5%. Even the coarser MMFF- and UFF-generated geometries, while less effective, still contributed to a notable MAE reduction of about 24%.

As depicted in Fig. 3d, we meticulously assessed the performance trajectory of DGT(3D) when pretrained on subsets of the PCQM4Mv2 dataset[36] of increasing size (0, 10K, 100K, and 1M molecules) for predicting $\varepsilon_{\mathrm{HOMO}}$ and $\varepsilon_{\mathrm{LUMO}}$. Enlarging the pretraining dataset delivered a monotonic reduction in MAE, underscoring the profound benefit of large-scale pretraining in facilitating DGT(3D) to learn a more generalised molecular representation for relevant downstream tasks. Quantitatively, the ultimate prediction MAEs for $\varepsilon_{\mathrm{HOMO}}$ and $\varepsilon_{\mathrm{LUMO}}$ were as low as 0.0302 eV and 0.0231 eV, respectively, corresponding to considerate improvements of 9.3% and 7.2% attributable to pretraining.

Fig. 3e compares DGT(3D) against some established state-of-the-art models designed for 3D molecular learning. The benchmark models included SchNet[37], DimeNet[38], SphereNet[39], SE(3) Transformer[40], GraphMVP[30], and 3DInfomax[41]. SchNet, DimeNet, and SphereNet represented 3D GNNs that use rotationally invariant features to predict scalar molecular properties. The SE(3) Transformer was included as a typical equivariant GNN that is robust to 3D roto-translations of molecules. GraphMVP and 3DInfomax were models adopting contrastive learning to enrich 2D graph structure embeddings with 3D conformational information. DGT(3D), whether pretrained or not, outperformed all other evaluated models, firmly establishing it as a highly effective 3D machine learning architecture for molecular property prediction. These results highlight the potential of leveraging 3D information in the dual graph representation to enhance the self-attention mechanism in molecular machine learning.

Attention-based interpretation

The global attention score matrices of DGT reveal the intricate interplay of molecular features for molecular property prediction, unravelling insights into the model's decision-making and identifying key contributing molecular substructures. To jointly account for the model's focus and the sensitivity of its prediction, the multi-head attention scores are multiplied with the absolute gradients of the prediction with respect to each element[42,43]. The resulting dot products qualify the significance values of pairwise relationships for the specific task, which are subsequently aggregated to evaluate the contribution of atoms and bonds. Fig. 4 presents molecular examples illustrating this attention-based interpretation across benchmark datasets.

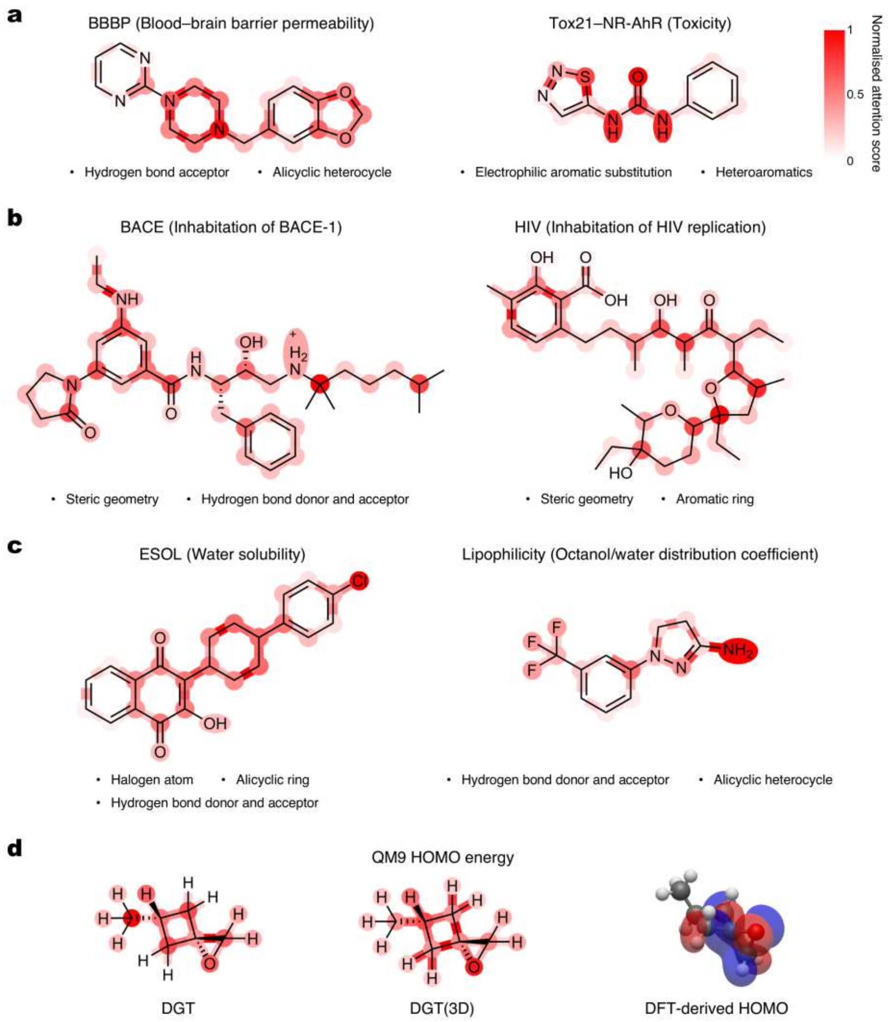

Fig. 4: Attention-based interpretation analysis results. a–d, Attention-based interpretation is conducted on molecular property prediction tasks across all four scientific domains: physiology (**a**), biophysics (**b**), physical chemistry (**c**), and quantum mechanics (**d**). Attention scores calculation concurrently considers the attention score matrices of atoms and bonds, as well as the absolute gradients of the prediction with respect to them. Darker colours denote a higher level of attention score assigned to the atom or bond. Attention scores calculated using DGT(3D) are provided for the QM9 HOMO energy prediction example, along with the DFT-derived HOMO at the B3LYP/6-31G(2df,p) level of theory.

Within the physiology domain, hydrogen bond acceptor atoms and alicyclic heterocycles are highlighted in the example molecule from the BBBP dataset (Fig. 4a). This observation aligns with established principles governing blood–brain barrier penetration, which generally favour molecules with high lipophilicity and a limited number of hydrogen bond donors and acceptors[44]. For the Tox21–NR-AhR toxicity prediction task, ligands of aryl hydrocarbon receptor, which is a transcription factor, are predominantly electron-deficient aromatic systems, closely correlated with the attentive region identified in the illustrated molecule[45].

The BACE and HIV datasets are representative for examining the attention-based interpretation in the biophysics domain. In both cases, attention weights are predominantly focused on atoms with steric geometries, with secondary attention distributed across hydrogen-bonding sites and aromatic rings for predicting the inhibition of human β-secretase 1[46] and HIV replication[47], respectively (Fig. 4b). These structural factors are widely recognised as determinants of inhibitor binding at target active sites.

Fig. 4c showcases the calculated attention scores for water solubility and lipophilicity predictions, two properties that are generally inversely correlated[48]. Hydrogen bond donors and acceptors, polar groups, and hydrophobic surface area are vital factors for them, to which the highlighted molecular substructures in the shown molecules conform.

In the QM9 HOMO energy prediction task, 3D information helps disentangle key molecular sections when incorporated into the DGT model[49]. Compared to DGT, DGT(3D) performs better to distribute greater attention to the epoxide and cyclobutane rings in the illustrated molecule, which is corroborated by the DFT-derived HOMO (Fig. 4d). These findings underscore the potential of DGT to offer valuable insights for molecular property at the molecular structural level.

Discussion

In this work, we present DGT as a graph transformer model which harnesses the dual graph representation for molecular property prediction. By enriching information in the molecular representation, the potential of the self-attention mechanism within the graph transformer model is unlocked for learning on molecules. DGT achieves state-of-the-art performance across 58 tasks from 10 datasets across four domains of the MoleculeNet benchmark. Further investigation reveals the effectiveness of the dual graph representation, graph-specific encodings, and 3D information incorporation for enhancing the prediction accuracy, as well as the attention mechanism's ability to identify molecular substructures critical to the target molecular property.

Deep learning has emerged as a powerful approach to expedite the exploration of molecules with desired properties across versatile domains. The graph transformer, which enables global dynamic node–node interactions regardless of their graph distances, is attracting wide interests in the field. Our work underscores the rationale to bridge the gap between molecular representation and the powerful attention mechanism for molecular property prediction. However, despite that DGT demonstrates massive benefits from utilising 3D information in molecular machine learning, the potential to generate high-fidelity molecular spatial structure is yet to be explored. Such structure generation model can be added to the current workflow of DGT as an end-to-end approach. In addition, some recent publications report improved pretraining on molecules using node-level context prediction over the graph-level attribute prediction[29]. DGT also holds the potential to promote more advanced domains such as conformation generation, molecular generative modelling, and chemical reaction prediction.

Looking ahead, DGT is poised to dedicate to high-throughput molecular screening and discovery, driven by the growing demand for efficient exploration of vast chemical spaces and rapid identification of novel compounds with tailored properties. We foresee DGT advancing deep learning on molecules and constituting a component of the upcoming AI-driven scientific research, improving both the prediction accuracy and interpretability of molecular properties.

Methods

Datasets and splitting strategy

We thoroughly evaluated DGT across four domains for molecular property prediction, encompassing ten datasets and a total of 58 subtasks from the MoleculeNet benchmark[26]. Datasets included in the physiology were BBBP, ClinTox, and Tox21, which comprises 12 classification tasks ranging from NR-AR to SR-p53, as well as SIDER, which covers 27 tasks related to adverse drug reactions. BACE and HIV were selected from the biophysics domain to assess classification performance on drug activity and antiviral screening, respectively; For physical chemistry, three regression tasks were employed: ESOL, FreeSolv, and Lipophilicity. Finally, the quantum chemistry domain was represented by the QM9 dataset, comprising 12 regression tasks targeting molecular quantum properties, ranging from basic properties such as dipole moments to electronic properties including frontier orbital energies. Randomly sampled subsets of the PCQM4Mv2 dataset were utilised as the training set for investigating the pretraining of DGT(3D).

We employed the scaffold split[50] method to partition datasets from the MoleculeNet benchmark into training, validation, and test sets with an 80:10:10 ratio. In this approach, splitting is based on the molecular scaffold—essentially the core structure of a molecule, essentially the fundamental ring system or backbone without any substituent groups. Compared to the random split, the scaffold split is more challenging by ensuring that the test set contains molecules with scaffolds that were not encountered during training, thereby requiring the model to generalise to new chemical structures. To ensure a fair comparison between DGT and baseline models, the model training in this work was repeated four times with the same data split to get the average result. The scaffold split method was implemented following the publicly available codes from the chemprop GitHub repository (https://github.com/aamini/chemprop.git).

Physiology. The BBBP dataset comprises 2,039 molecules annotated with binary labels indicating whether each compound can penetrate the blood–brain barrier, a key consideration in central nervous system drug development. The ClinTox dataset contains 1,478 compounds categorised according to whether they have been approved or withdrawn from clinical trials due to severe toxicity. Tox21 is a public database consisting of 7,831 instances qualitatively evaluated across 12 biological targets, including nuclear receptors and stress response pathways, providing a diverse set of multi-label toxicity prediction tasks. The SIDER dataset maps 1,427 marketed drugs to 27 types of clinically reported adverse drug reactions, supporting the association between molecular structures and side-effect profiles for developing safer pharmaceuticals.

Biophysics. The BACE dataset consists of 1,513 small-molecule inhibitors annotated for their binding activity toward the human β-secretase 1 enzyme, a critical target in Alzheimer's disease research. With 41,127 compounds, the HIV dataset provides profound data on molecules' ability to inhibit HIV replication, offering a large-scale benchmark for antiviral activity classification.

Physical chemistry. The ESOL dataset collects 1,128 molecules with experimentally measured aqueous solubility values, supporting the investigation of how molecular features influence solubility. The FreeSolv dataset is a curated collection of 642 small organic compounds with hydration free energies, serving as a data source for studying drug solubility and stability. Lipophilicity, which describes how well a substance interacts with non-polar environments, is captured in a dataset with 4,200 compounds with experimental octanol–water partition coefficients, offering essential data for analysing the pharmacokinetic behaviour.

Quantum Mechanics. The quantum mechanics domain is represented by the QM9 dataset, comprising 133,885 small molecules composed of up to nine heavy atoms (C, O, N, and F), each annotated with quantum mechanical properties computed at the B3LYP/6-31G(2df,p) level of theory. The dataset includes 12 regression targets that capture fundamental quantum molecular properties crucial for versatile domains. These properties encompass dipole moment, polarisability, squared radius, zero-point vibrational energy, heat capacity at constant volume, highest occupied molecular orbital (HOMO) energy, lowest unoccupied molecular orbital (LUMO) energy, HOMO–LUMO energy gap, internal energy at 0 K, internal energy at standard state, enthalpy, and Gibbs free energy.

Experimental settings

The experimental setting section consists of evaluation metrics and pretraining setup.

Evaluation metrics. We benchmarked DGT across 58 molecular property prediction tasks spanning four distinct domains, each evaluated using domain-appropriate metrics. The AUC-ROC was utilised for classification tasks within the physiology and biophysics domains, as it quantifies a model's ability to distinguish between classes across various threshold settings; a higher AUC-ROC value indicates better classification performance. For physical chemistry, the RMSE was adopted to assess the difference between predicted and true values. MAE was employed for the quantum mechanics domain as a routine metric to measure the average magnitude of prediction errors.

Pretraining setup. We investigated the applicability of the pretraining strategy on DGT(3D) using the large-scale PCQM4Mv2 dataset[36]. In this setup, DGT(3D) was first pretrained on the PCQM4Mv2 dataset to predict the HOMO–LUMO energy gap, a relevant task intended to enhance the downstream prediction of HOMO and LUMO energies within the QM9 dataset. To assess the impact of the pretraining dataset size, we carried out a series of pretraining runs using subsets with 10K, 100K, and 1M molecules randomly sampled from PCQM4Mv2. DGT(3D) was pretrained on these sampled molecules for 100 epochs incorporating a linear warmup strategy over the first 5 epochs, using the AdamW optimiser with a batch size of 128 and a small learning rate of 0.0002, scheduled via cosine annealing. The resulting pretrained weights were then utilised to initialise DGT(3D) for training on the QM9 dataset. Specifically, PCQM4Mv2 involves a total of 22 elements, covering those present in the QM9 dataset (H, C, O, N, and F); relevant atomic embeddings were accordingly extracted and concatenated to initialise the atom embedding matrix for QM9.

Self-attention modules

For molecular property prediction, DGT encodes tuples $\mathrm{G}_a = (\mathrm{N}_a, \mathrm{E}_a)$ and $\mathrm{G}_b = (\mathrm{N}_b, \mathrm{E}_b)$ as both atom and bond graphs, respectively, where N_a indicates the set of atoms, $\mathrm{E}_a \subseteq \mathrm{N}_a \times \mathrm{N}_a$ the set of directed bonds, N_b the set of undirected bonds, and $\mathrm{E}_a \subseteq \mathrm{N}_b \times \mathrm{N}_b$ the set of neighbour pairs of undirected bonds. Numbers of atoms and bonds are denoted with $N_a = |\mathrm{N}_a|$ and $N_b = |\mathrm{N}_b|$, respectively. Given the atom feature dimension size as d_a, the atom features are encoded as rows in a matrix $\mathbf{N}_a \in \mathrm{R}^{N_a \times d_a}$; likewise, the bond features are encoded as $\mathbf{N}_a \in \mathrm{R}^{N_b \times d_b}$, and d_b is the bond feature dimension size.

The biased multi-head attention mechanism calculation can be formulated as:

$$a_{i,j} = \exp\left(\frac{Q_i \cdot K_j}{\sqrt{d_k}} + E_{i,j}^{att}\right)$$

$$s_{i,j} = a_{i,j} \Big/ \sum_j a_{i,j}$$

$$h_i = \sum_j s_{i,j}\left(V_j + E_{i,j}^{val}\right)$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are respective projections of $\mathbf{X}$ ($\mathbf{N}_a$ or $\mathbf{N}_b$) for query, key, and value; $\mathbf{E}^{att}$ and $\mathbf{E}^{val}$ are respective projections of the node–node encoding $\mathbf{E}$ ($\mathbf{E}_a$ or $\mathbf{E}_b$) to attention bias and edge feature injection. d_k, which denotes dimensionality of the key vectors, is involved for stabilising the training process. h_i is the hidden vector for node i.

The node–node encoding $\mathbf{E}$ is the sum of encodings of relative position and ring structure. Notably, DGT(3D) enables incorporating bond lengths to the node feature matrix $\mathbf{N}_b$, atom–atom distances to $\mathbf{E}_a$, and bond–bond angles to $\mathbf{E}_b$. To inform the model with enriched local geometric representation, interatomic distances are expanded using the physics-inspired orthogonal Bessel basis functions to improve expressivity and stability; bond angular information is encoded with spherical harmonics functions defined on the unit sphere to ensure rotational consistency[51]. Such geometry-aware representations are crucial for accurate property prediction.

With this attention calculation hereafter referred to as $\mathrm{MHA}(\mathbf{X}, \mathbf{E}^{att}, \mathbf{E}^{val})$, the DGT layer can be given as:

$$\overline{\mathbf{H}} = \mathrm{BatchNorm}(\mathbf{X} + \mathrm{MLP}(\mathrm{MHA}(\mathbf{X}, \mathbf{E}^{att}, \mathbf{E}^{val})))$$

$$\overline{\mathbf{X}} = \mathrm{BatchNorm}(\mathrm{FFN}(\overline{\mathbf{H}}))$$

Here, the input hidden node feature matrix $\mathbf{X}$ is added to $\mathrm{MHA}(\mathbf{X}, \mathbf{E}^{att}, \mathbf{E}^{val})$ transformed by the multilayer perceptron (MLP). Feed forward network (FFN) follows to implement nonlinear transformations to each token embedding to obtain the output node features matrix $\overline{\mathbf{X}}$. The batch normalisation (BatchNorm) is applied to avoid overfitting.

For the readout part, atom and bond features are globally averaged and concatenated for predicting the molecular property as $\mathrm{MLP}(\mathrm{GAP}(\mathbf{X}_a) \Vert \mathrm{GAP}(\mathbf{X}_b))$, where GAP is the global average pooling operation and the Rectified Linear Unit activation function is attached to linear layers of MLP.

Baselines

DGT was comprehensively compared with state-of-the-art machine learning methods on datasets from the MoleculeNet benchmark. Random forest (RF) is adopted as a baseline machine learning model[26]. We also compared DGT with some advanced graph neural network (GNN) models, including Attentive FP[27], D-MPNN[28], pretrained GNN[29], GraphMVP[30], MolCLR[31], and MoleBERT[32]. GROVER presented as a representation model for the graph transformer[33]. UniMol was included as a representation of transformer-based 3D geometric deep learning models for molecular property prediction[34]. Performances of GraphMVP, MolCLR, MoleBERT, GROVER, and UniMol are derived from ref.[52] with the same data splitting. Attentive FP and D-MPNN is implemented using the DeepChem package (https://github.com/deepchem/deepchem.git), random forest is based on scikit-learn (https://github.com/scikit-learn/scikit-learn.git), and pretrained GNN is reproduced according to https://github.com/snap-stanford/pretrain-gnns.git.

In detail, RF employs ECFP4 as input features—a widely adopted circular fingerprint in cheminformatics and molecular machine learning, which encodes molecules by capturing local atom-centred substructures. Attentive FP extends conventional GNNs by incorporating the attention mechanism that dynamically adjusts the importance weights of neighbouring atoms and bonds during message passing. D-MPNN refines standard message passing neural networks by directing messages over bonds instead of atoms, which improves the encoding of bond-level interactions critical for accurate molecular property prediction. The pretrained GNN baseline demonstrates the performance gains of a GNN architecture by informing it with large-scale molecular datasets before fine-tuning on downstream tasks. GraphMVP leverages a multi-view contrastive learning strategy, aligning information from the 2D molecular graph and its corresponding 3D conformer to encourage the model to learn representations consistent across different structural modalities.

MolCLR further advances contrastive learning by generating diverse graph augmentations to improve the robustness of learnt features against input perturbations. MoleBERT adapts language model pretraining principles to molecular graphs by introducing masked atom and bond prediction tasks, facilitating the learning of versatile graph-level encoders that can generalise across molecular prediction tasks. GROVER combines self-supervised learning objectives with a graph transformer backbone, learning rich contextualised molecular representations from millions of unlabelled molecules.

For 3D molecular learning, DGT(3D) was benchmarked against SchNet[37], DimeNet[38], SphereNet[39], and SE(3) Transformer[40], besides GraphMVP[30] and 3DInfomax[41] that leverage 3D information to enrich 2D molecular embedding. SchNet represents one of the earliest deep learning architectures explicitly designed for molecules and materials, incorporating continuous-filter convolutional layers to directly model quantum interactions. By treating interatomic distances as continuous inputs, SchNet effectively captures long-range dependencies for predicting quantum mechanical properties. Based on that, DimeNet introduces directional message passing by encoding angular information between bonded atoms, thereby enabling the model to learn higher-order geometric interactions. SphereNet extends this paradigm by leveraging spherical message passing, integrating both angular and radial basis functions to achieve a more complete characterisation of local geometric environments. The SE(3) Transformer generalises attention mechanisms to 3D Euclidean space by enforcing equivariance under SE(3) group transformations, allowing the model to learn molecular representations invariant to rotations and translations while preserving geometric symmetries. Packages for implementing these compared models are as follows: SchNet and DimeNet, https://github.com/pyg-team/pytorch_geometric.git; SphereNet, https://github.com/divelab/DIG.git; SE(3) Transformer, https://github.com/chao1224/Geom3D.git.

Data availability

Datasets used for benchmarking the DGT model are available at https://moleculenet.org/datasets. The PCQM4Mv2 dataset was used in pretraining DGT(3D), which is accessible via https://ogb.stanford.edu/docs/lsc/pcqm4mv2. Source data are provided with this paper.

Code availability

All codes are available via GitHub at https://github.com/zhangsy-ryan/DGT.git.

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S.Y.Z. conceived the project and contributed to the technical implementation. A.A.L. provided project administration and acquired funding. S.Y.Z. wrote the original paper. All authors reviewed and edited the manuscript.

Competing interests

The authors declare no competing interests.

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